Mechanical Behaviour of Engineering Materials

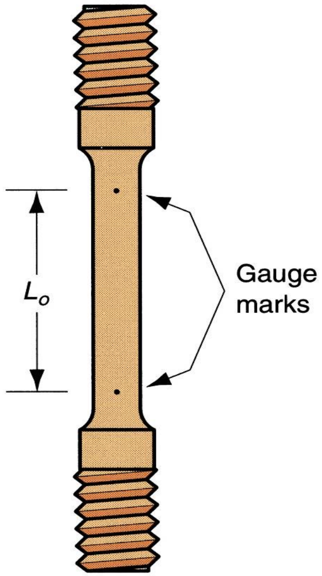
Groover: Chapter 3

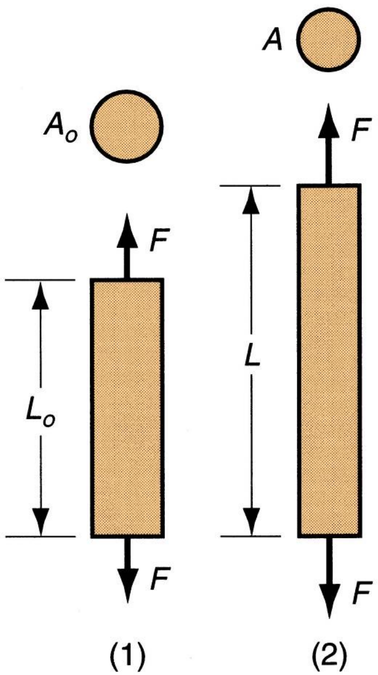
Learning Targets

The student:

- knows the basic mechanical properties of materials;
- knows how to test supplied materials.

TENSILE TEST

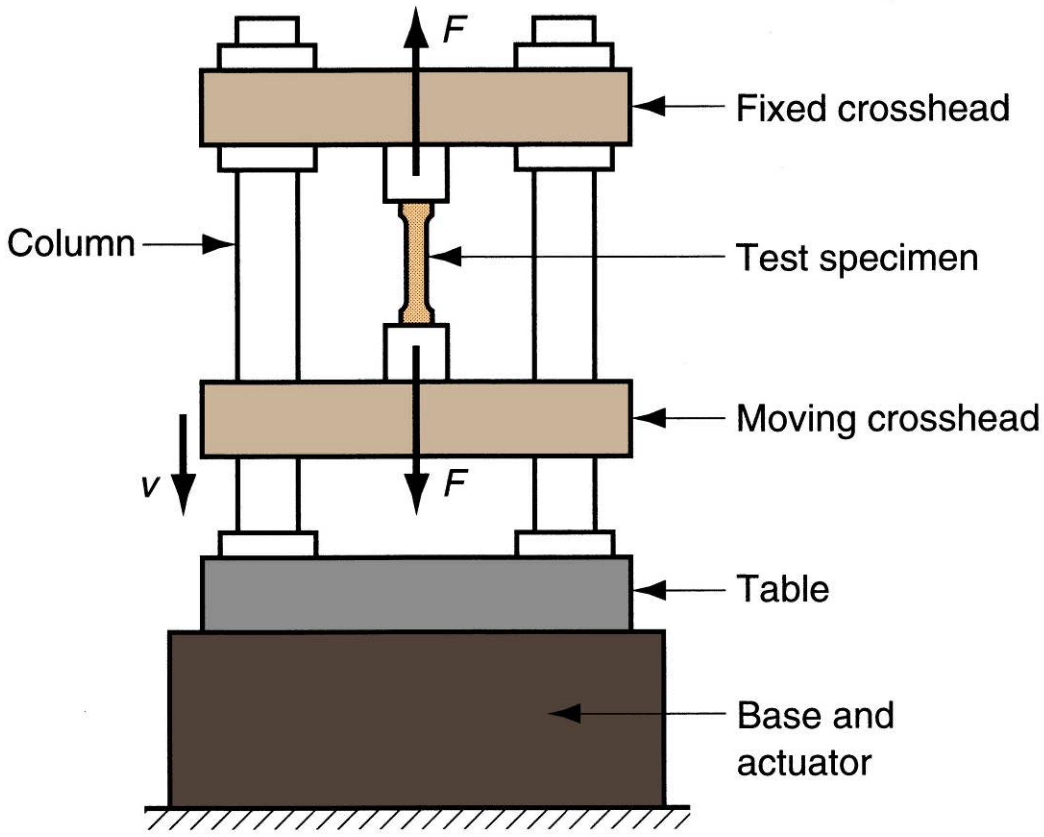




Standards: UNI EN 10002, ISO 204, ASTM: E646

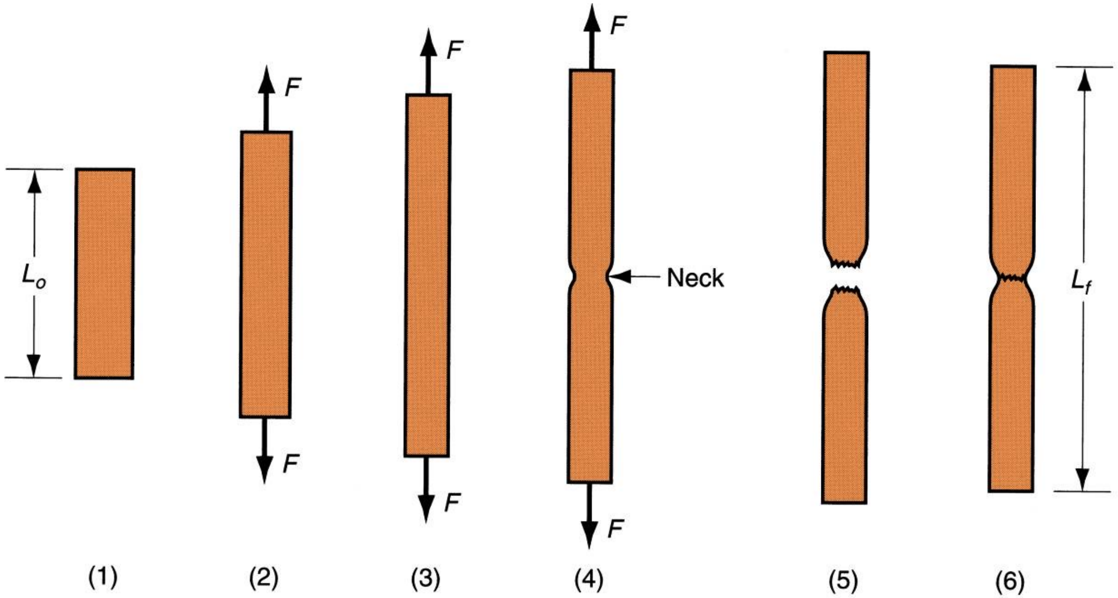
Tensile test: tensile force applied in (1) and (2) resulting elongation of material

TENSILE TEST SETUP



TENSILE TEST SEQUENCE

Typical progress of a tensile test: (1) beginning of test, no load; (2) uniform elongation and reduction of cross-sectional area; (3) continued elongation, maximum load reached; (4) necking begins, load begins to decrease; and (5) fracture. If pieces are put back together as in (6), final length can be measured.



Engineering Stress & Strain

Engineering stress

$$s = \frac{F}{A_0}$$

Where:

s = engineering stress [MPa]

F = applied force [N]

A_0 = original area of test specimen [mm^2]

Engineering strain

$$e = \frac{L - L_0}{L_0}$$

Where:

e = engineering strain

L = actual length

L_0 = original length

True Stress and True Strain

True Stress: Stress value obtained by dividing the instantaneous area into applied load.

$$\sigma = \frac{F}{A}$$

Where:
σ = true stress [MPa]
F = force [N]
A = actual (instantaneous) area [mm^2]

True Strain: Provides a more realistic assessment of "instantaneous" elongation per unit length.

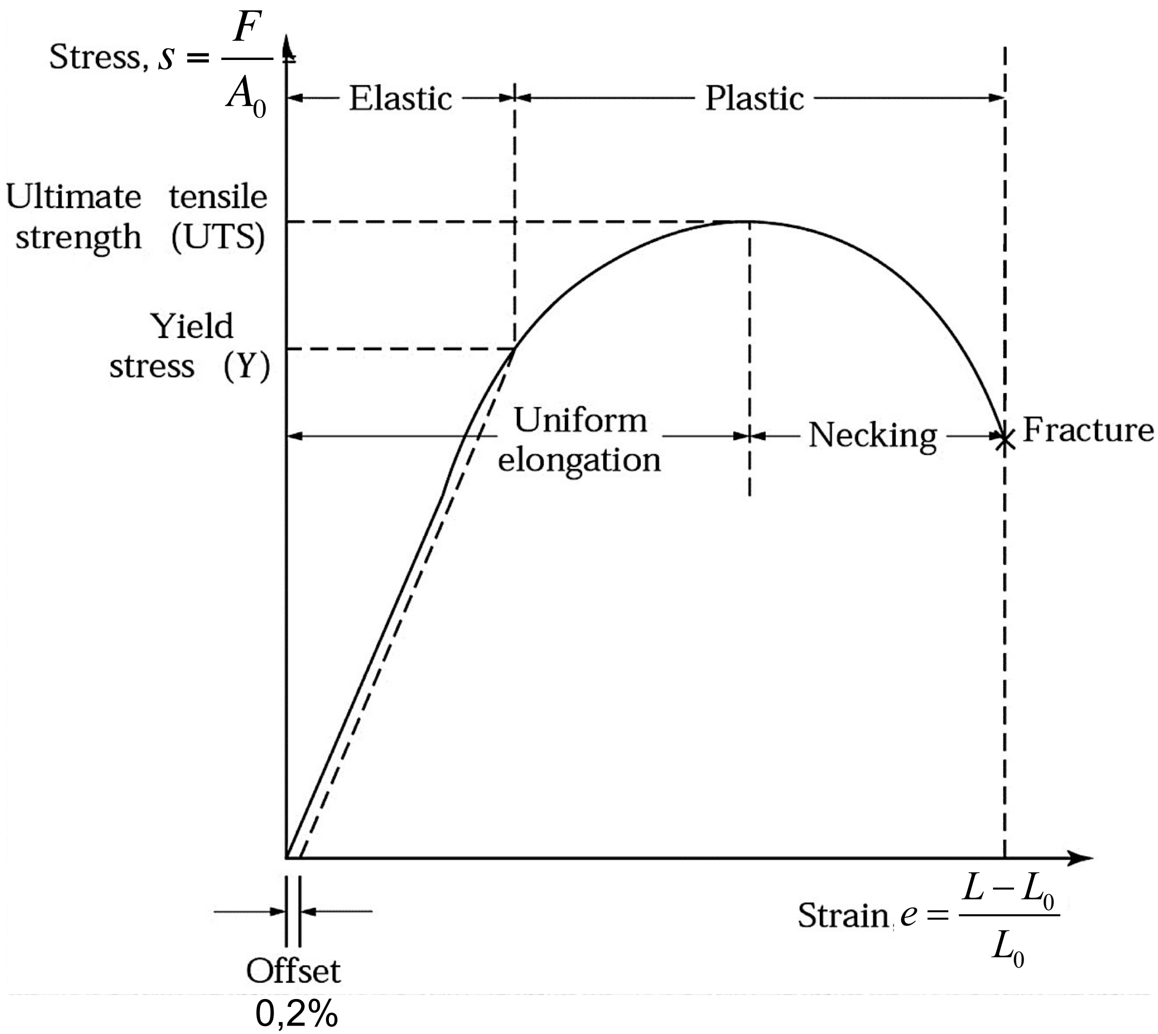
$$d\varepsilon = \frac{dl}{l}$$

$$\varepsilon = \int_{L_0}^{L} \frac{dl}{l} = \ln\left(\frac{L}{L_0}\right)$$

Where:
ε = true strain
L = actual length
L_0 = original length

ENGINEERING STRESS-STRAIN PLOT

Typical engineering stress-strain plot in a tensile test of a metal.



Elastic Region

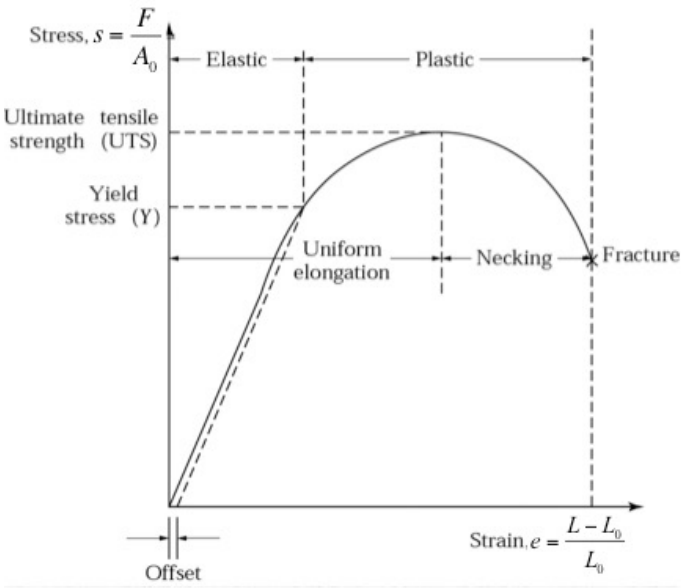
- Relationship between stress and strain is linear.
- Material returns to its original length when stress is removed (elastic recovery).
- ***Hooke's Law***: $s = E\,e$
  - where E is the modulus of elasticity, or Young's modulus (after T. Young 1773-1829);
  - E is a measure of the inherent stiffness of a material. Its value differs for different materials.

YIELD POINT

As stress increases, a point in the stress-strain curve is finally reached when the material begins to yield.

Yield point Y is the point of change from elastic to plastic behaviour of the material.

Y : a strength property

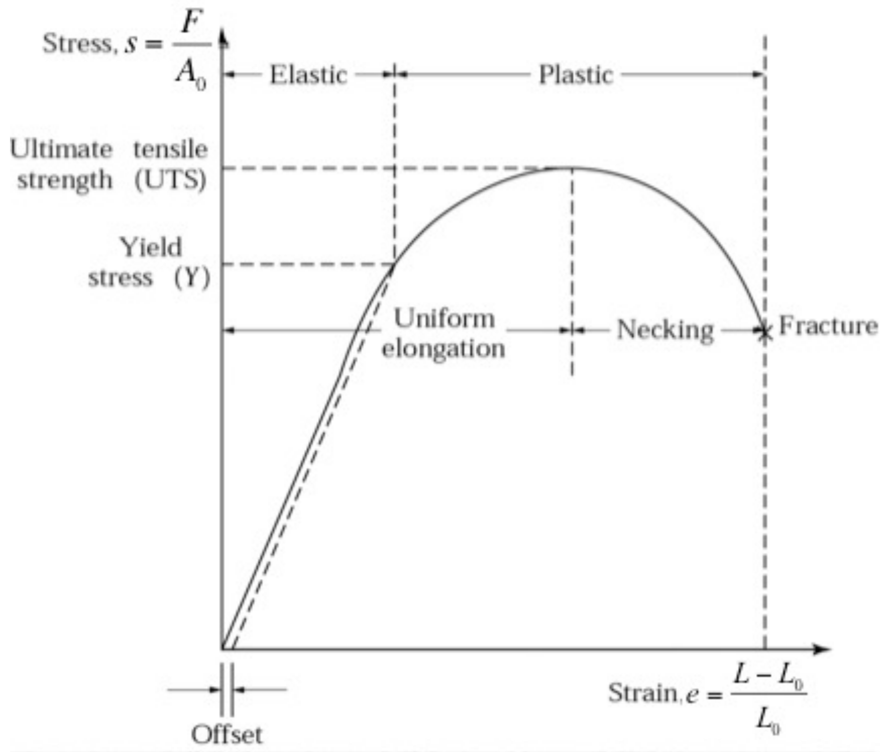


Other names for yield point:

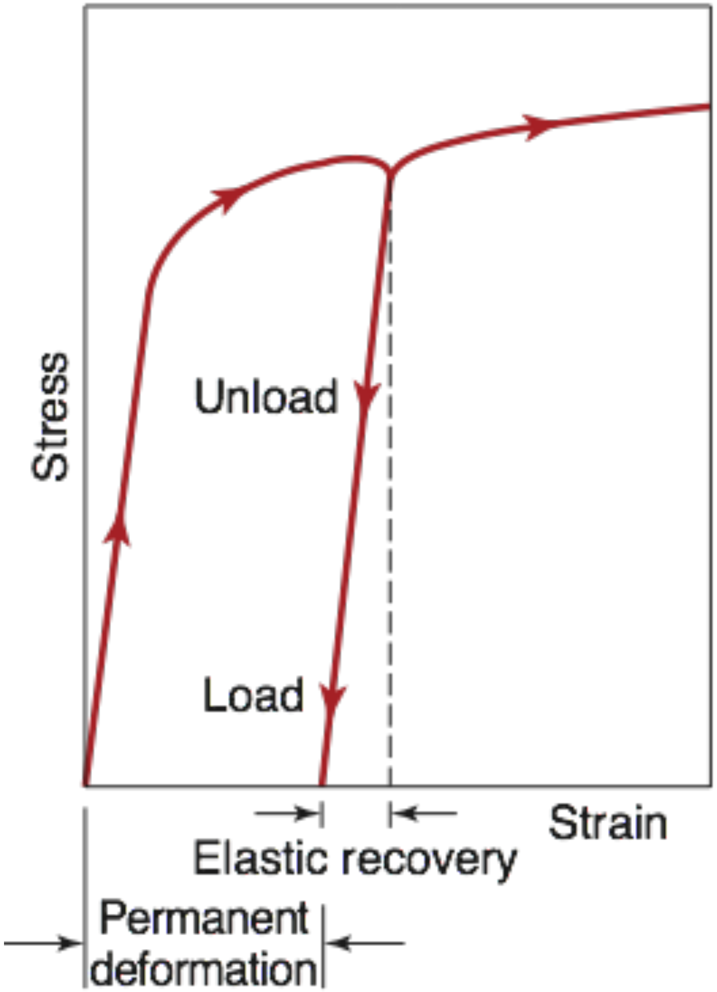
yield strength, yield stress, and elastic limit

Plastic Region

- Yield point marks the beginning of plastic deformation.
- As load is increased beyond *Y*, elongation proceeds at a much faster rate than before, causing the slope of the curve to change dramatically.



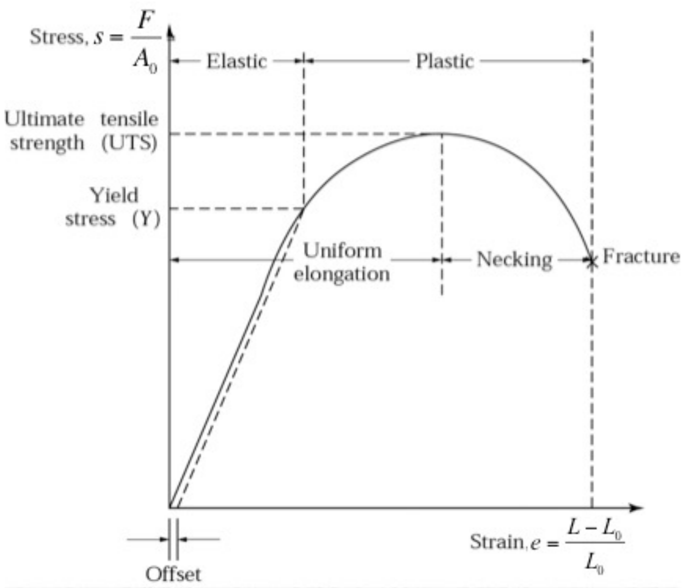
LOADING & UNLOADING



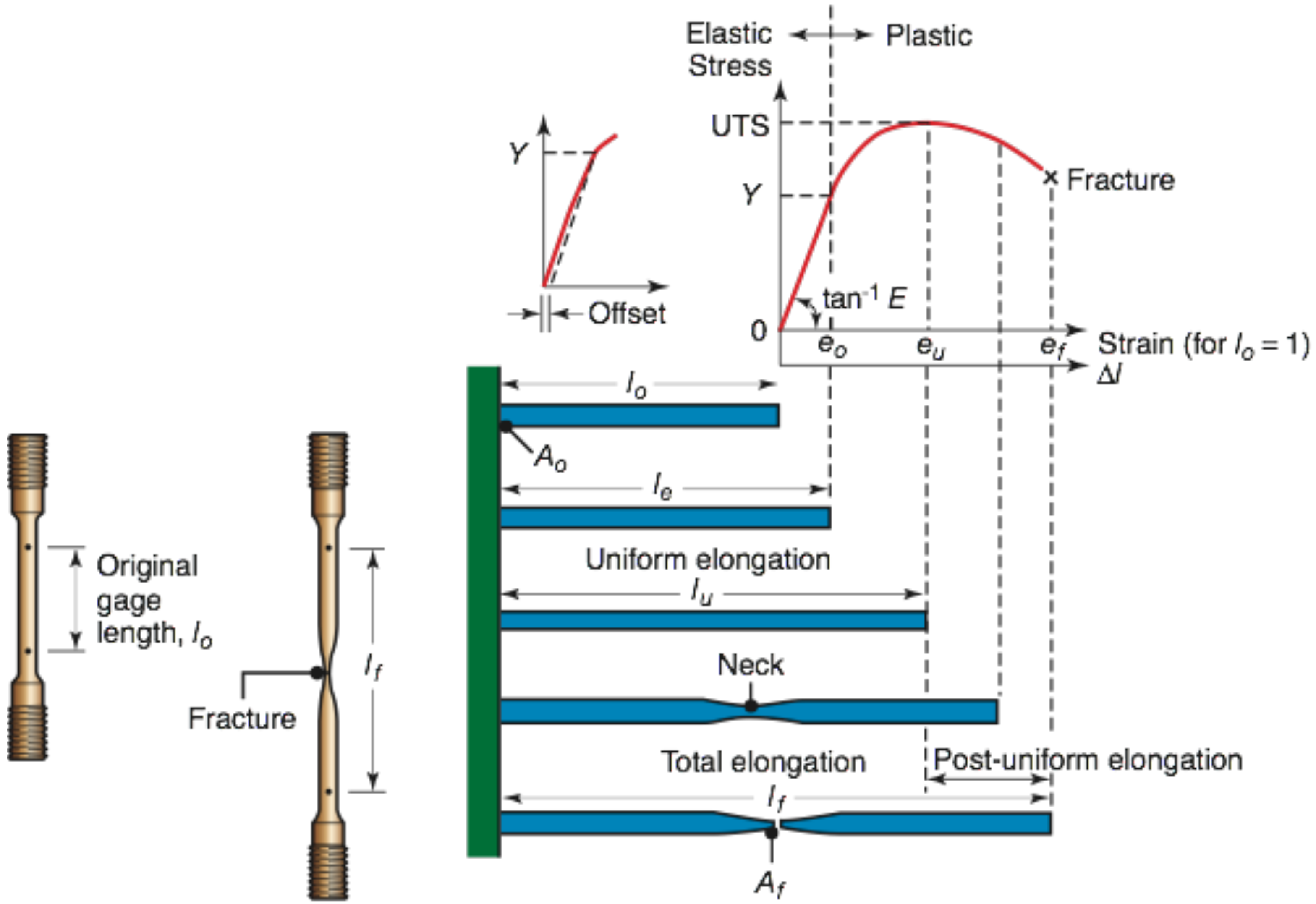
ULTIMATE TENSILE STRENGTH

- Elongation is accompanied by a uniform reduction in cross-section, consistent with maintaining constant volume.
- Finally, the applied load *F* reaches a maximum value, and engineering stress at this point is called the tensile strength or *UTS* (*Ultimate Tensile Strength*).

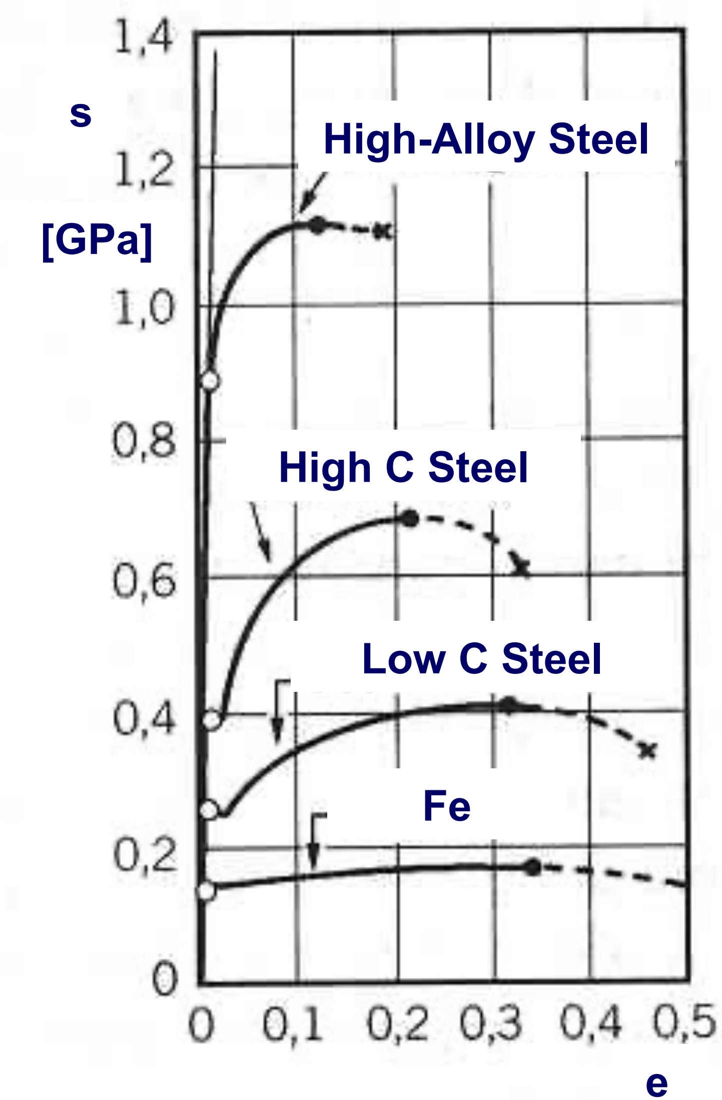
$$UTS = \frac{F_{\max}}{A_0}$$



NECKING


Elastic
Plastic
Stress
UTS
Y
Fracture
Y
Offset
tan⁻¹ E
0
e_o
e_u
e_f
Strain (for l_o = 1)
Δl
l_o
A_o
l_e
Uniform elongation
l_u
Original
gage
length, l_o
l_f
Fracture
Neck
Total elongation
Post-uniform elongation
l_f
A_f

Engineering Stress-Strain Plot



Flow Curve

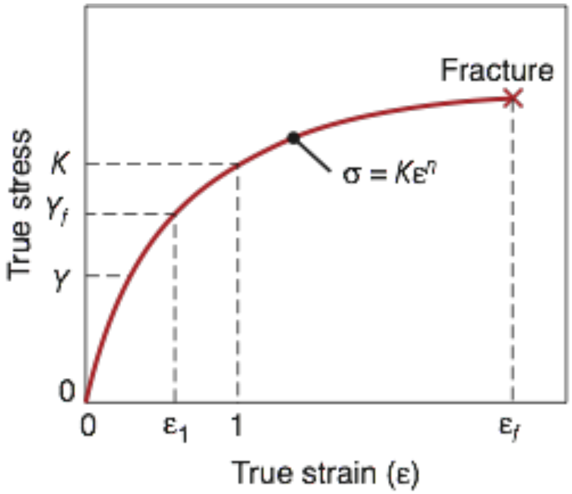
The relationship between true stress and true strain in the plastic region is represented by the so-called Flow Curve:

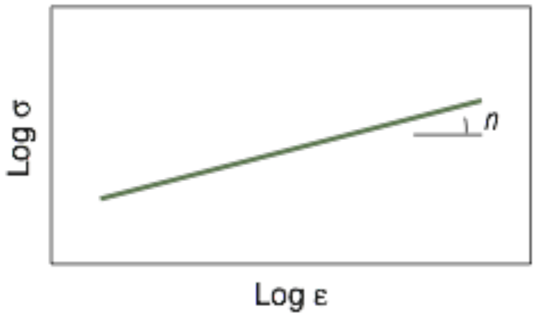
$$\sigma = K\varepsilon^n$$

Where:

K is the strength coefficient

n is the strain hardening exponent





STRAIN HARDENING

- Note that true stress increases continuously in the plastic region until necking.
- It means that the metal is becoming stronger as strain increases.
- This is the property called **strain hardening**.

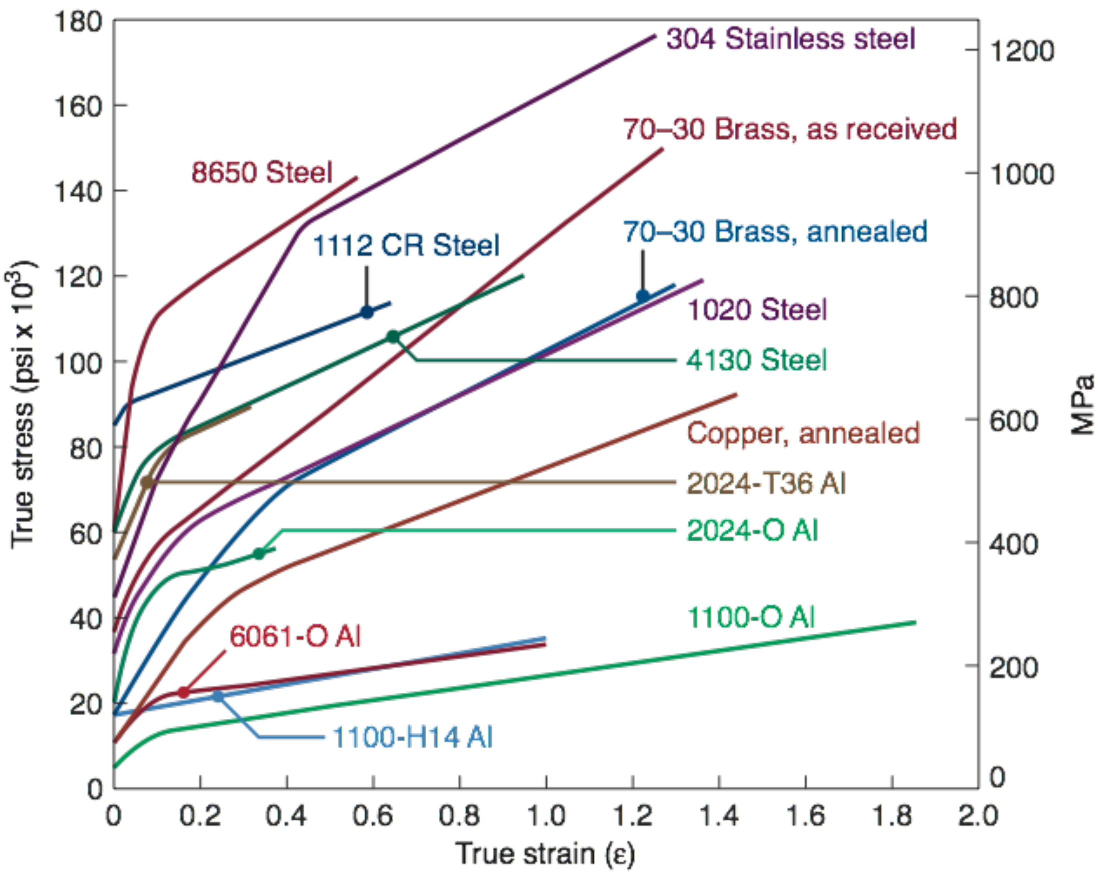
- In the engineering stress-strain curve, the significance of this is lost because stress is based on an incorrect area value.

FLOW CURVE

$$\sigma = K\varepsilon^n$$

Material	K (MPa)	n
Aluminum, 1100-O	180	0.20
2024-T4	690	0.16
5052-O	210	0.13
6061-O	205	0.20
6061-T6	410	0.05
7075-O	400	0.17
Brass, 7030, annealed	895	0.49
85-15, cold rolled	580	0.34
Bronze (phosphor), annealed	720	0.46
Cobalt-base alloy, heat treated	2070	0.50
Copper, annealed	315	0.54
Molybdenum, annealed	725	0.13
Steel, low carbon, annealed	530	0.26
1045 hot rolled	965	0.14
1112 annealed	760	0.19
1112 cold rolled	760	0.08
4135 annealed	1015	0.17
4135 cold rolled	1100	0.14
4340 annealed	640	0.15
17-4 P-H, annealed	1200	0.05
52100, annealed	1450	0.07
304 stainless, annealed	1275	0.45
410 stainless, annealed	960	0.10

Note: 100 MPa = 14,500 psi.



Categories of Stress-Strain Relationship

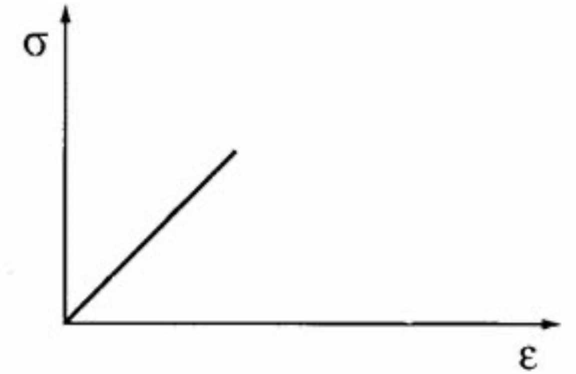
- Perfectly elastic
- Elastic and perfectly plastic
- Elastic and strain hardening

Stress-Strain Relationship: Perfectly Elastic

Behavior is completely defined by modulus of elasticity *E.*

Fractures rather than yielding to plastic flow.

Brittle materials: ceramics, many cast irons, and thermosetting polymers.



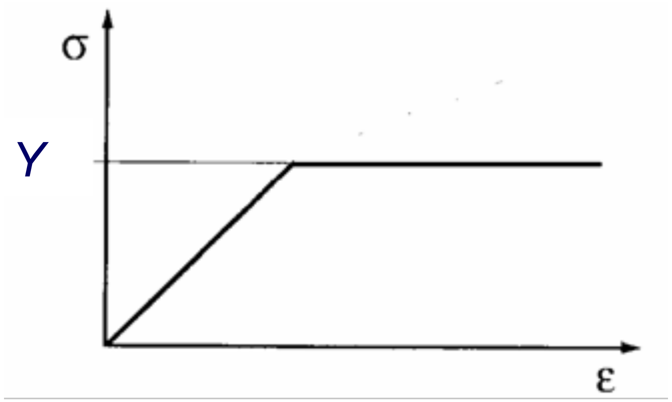
Stress-Strain Relationship: Elastic and Perfectly Plastic

Stiffness defined by modulus of elasticity *E*.

Once *Y* is reached, it deforms plastically at same stress level.

Flow curve: $K = Y$, $n = 0$

Metals behave like this when heated to sufficiently high temperatures (above recrystallization).

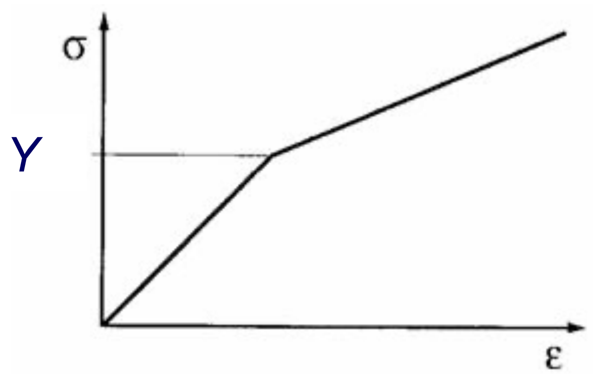


Stress-Strain Relationship: Elastic and Strain Hardening

Hooke's Law in elastic region, yields at *Y.*

Flow curve: $K > Y$, $n > 0$

Most ductile metals behave this way when cold worked.



DUCTILITY

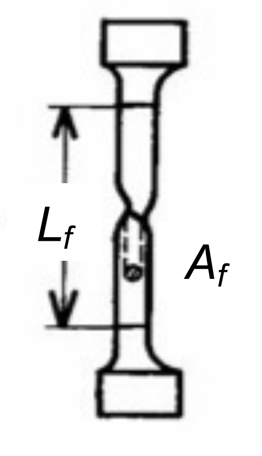
Ability of a material to plastically strain without fracture.

First Measure of Ductility = Total Elongation

$$EL = \frac{L_f - L_0}{L_0}$$

Where:

- EL = Total Elongation
- L_0 = original specimen length
- L_f = specimen length at fracture, it is measured as the distance between gage marks after the two pieces of the specimen are put back together



Ductility

Ability of a material to plastically strain without fracture.

Second Measure of Ductility = Area Reduction

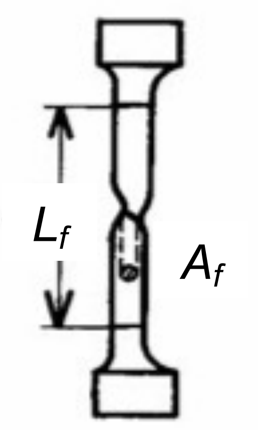
$$AR = \frac{A_0 - A_f}{A_0}$$

Where:

AR = Area Reduction

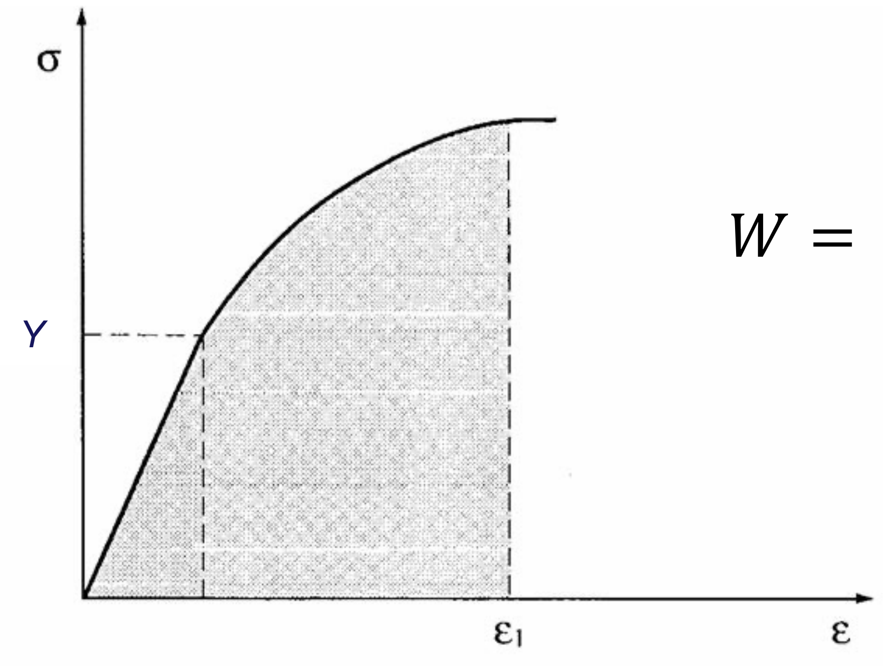
A_0 = original cross-sectional area of the test specimen

A_f = cross-sectional area of the test specimen at fracture



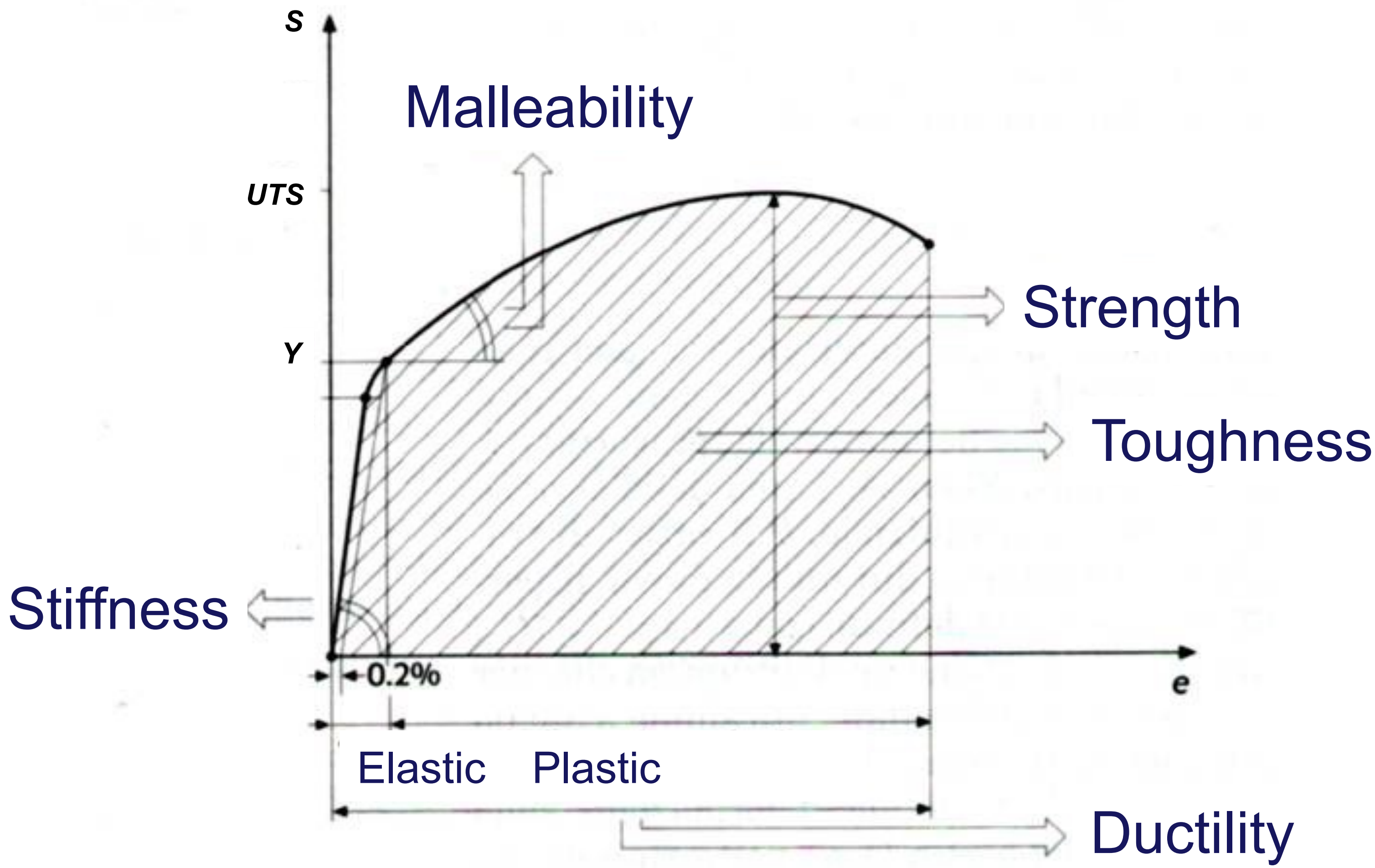
TOUGHNESS

Amount of energy per unit volume that the material dissipates prior to fracture.



$$W = \int_0^{\varepsilon_1} \sigma \cdot d\varepsilon$$

Mechanical Properties

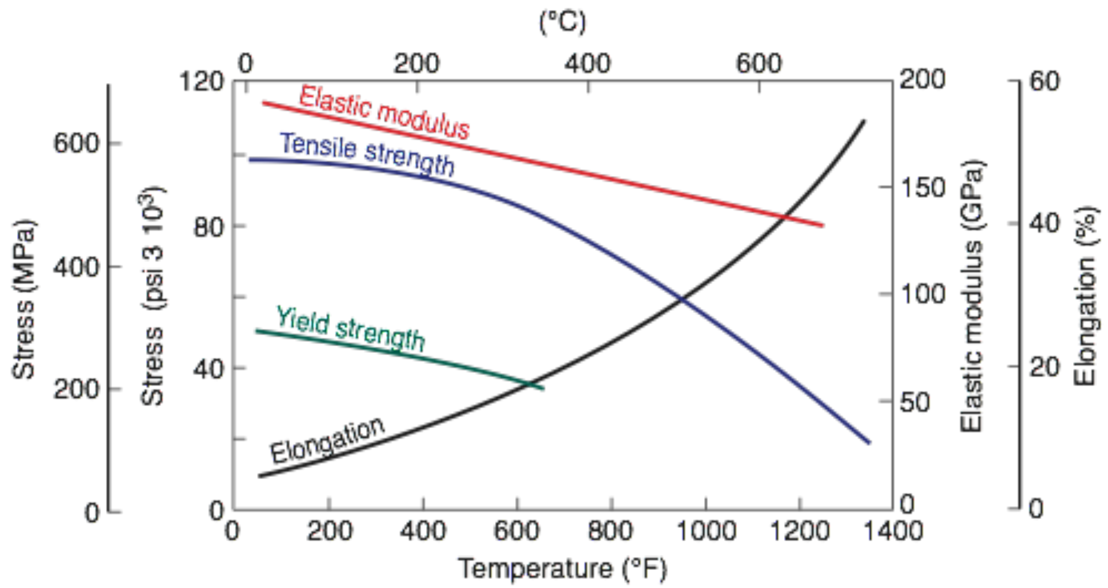


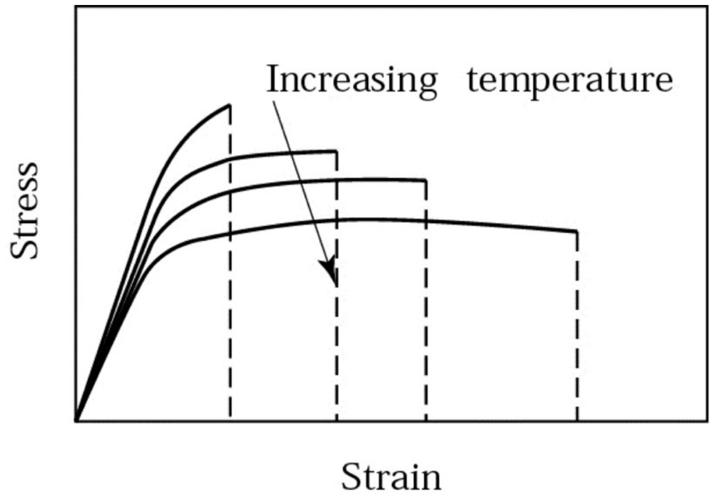
MECHANICAL PROPERTIES

	E (GPa)	Y (MPa)	UTS (MPa)	Elongation in 50 mm (%)	Poisson's Ratio (ν)
METALS (WROUGHT)					
Aluminum and its alloys	69-79	35-550	90-600	45-5	0.31-0.34
Copper and its alloys	105-150	76-1100	140-1310	65-3	0.33-0.35
Lead and its alloys	14	14	20-55	50-9	0.43
Magnesium and its alloys	41-45	130-305	240-380	21-5	0.29-0.35
Molybdenum and its alloys	330-360	80-2070	90-2340	40-30	0.32
Nickel and its alloys	180-214	105-1200	345-1450	60-5	0.31
Steels	190-200	205-1725	415-1750	65-2	0.28-0.33
Stainless steels	190-200	240-480	480-760	60-20	0.28-0.30
Titanium and its alloys	80-130	344-1380	415-1450	25-7	0.31-0.34
Tungsten and its alloys	350-400	550-690	620-760	0	0.27
NONMETALLIC MATERIALS					
Ceramics	70-1000	—	140-2600	0	0.2
Diamond	820-1050	—	—	—	—
Glass and porcelain	70-80	—	140	0	0.24
Rubbers	0.01-0.1	—	—	—	0.5
Thermoplastics	1.4-3.4	—	7-80	1000-5	0.32-0.40
Thermoplastics, reinforced	2-50	—	20-120	10-1	—
Thermosets	3.5-17	—	35-170	0	0.34
Boron fibers	380	—	3500	0	—
Carbon fibers	275-415	—	2000-5300	1-2	—
Glass fibers (S, E)	73-85	—	3500-4600	5	—
Kevlar fibers (29, 49, 129)	70-113	—	3000-3400	3-4	—
Spectra fibers (900, 1000)	73-100	—	2400-2800	3	—

TEMPERATURE EFFECT

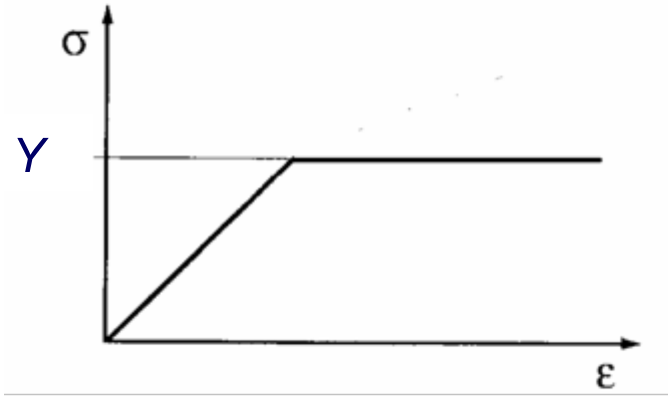
Most materials display similar temperature sensitivity for elastic modulus, yield strength, ultimate strength, and ductility.





Recrystallization in Metals

- Most metals strain harden at room temperature according to the flow curve ($n > 0$).
- But if heated to sufficiently high temperature and deformed, strain hardening does not occur:
  - Instead, new grains are formed that are free of strain;
  - The metal behaves as a perfectly plastic material, that is $n = 0$.



Recrystallization Temperature

- Formation of new *strain-free* grains is called recrystallization.
- Recrystallization temperature of a given metal is around 1/2 - 2/3 its melting point (0,5 – 0,67 T_m) as measured on absolute temperature scale [K].
- Recrystallization takes time - the recrystallization temperature is specified as the temperature at which new grains are formed in about one hour.

Recrystallization and Manufacturing

- Recrystallization can be exploited in manufacturing.
- Heating a metal to its recrystallization temperature prior to deformation allows a greater amount of straining, and lower forces and power are required to perform the process.
- Forming metals at temperatures above recrystallization temperature is called ***hot working***.

Hardness and Hardness Test

Hardness: Resistance to permanent indentation.

- Good hardness generally means material is resistant to scratching and wear.
- Most tooling used in manufacturing must be hard for scratch and wear resistance.

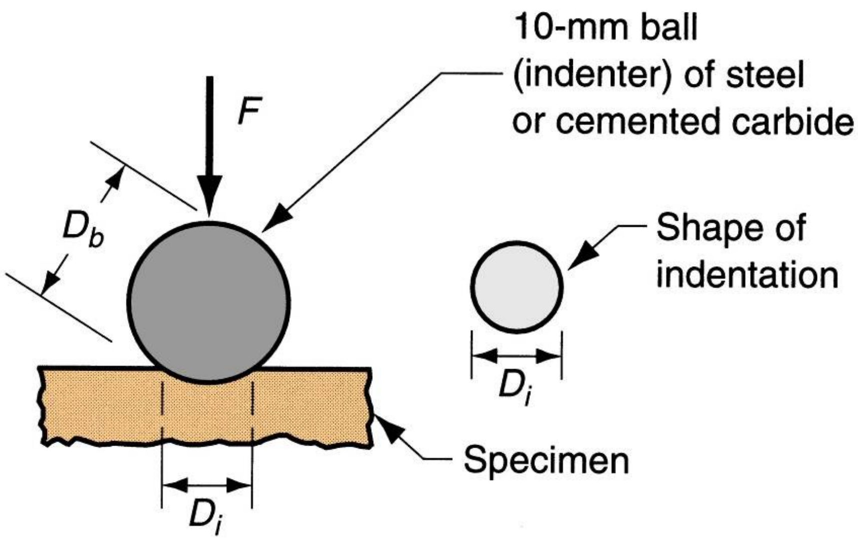
Variety of testing methods are appropriate due to differences in hardness among different materials.

- Most well-known hardness tests are Brinell, Vickers and Rockwell.
- Other test methods are also available, such as Knoop, Scleroscope, and durometer.

BRINELL HARDNESS TEST

Widely used for testing metals and nonmetals of low to medium hardness.

A hard ball (D_b = 10 mm) is pressed into specimen surface with a load (29400 N = 3000 kg) for 15 s.



(a) Brinell

Brinell Hardness Number (BHN) =

Load (*F*) divided into indentation area (*S*)

$$HB = 0{,}102 \cdot \frac{F}{S} = 0{,}102 \cdot \frac{2F}{\pi D_b (D_b - \sqrt{D_b^2 - D_i^2})}$$

Where:

HB = Brinell Hardness Number (BHN)

F = indentation load [N]

D_b = diameter of ball [mm]

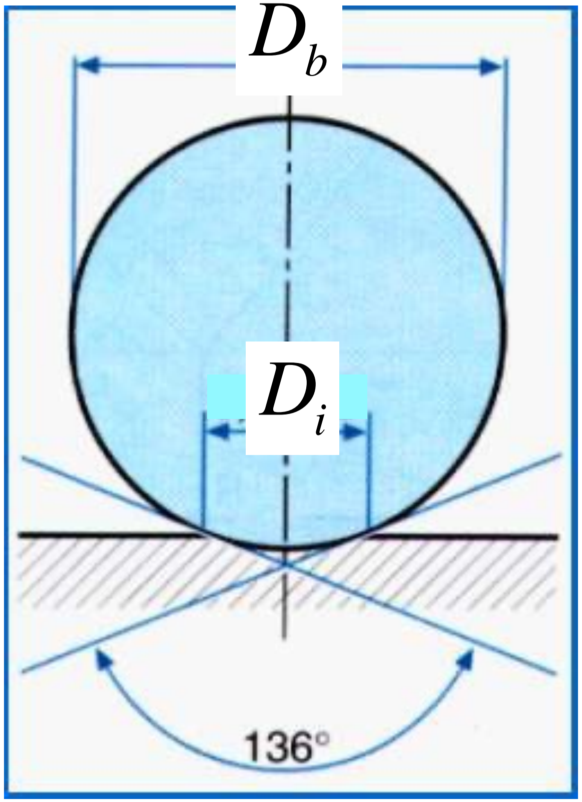
D_i = measured diameter of indentation [mm]

Brinell Hardness Test

To compare different tests, the indentation of the material has to be adequate.

Ideal Test $$\frac{D_i}{D_b} = \cos\frac{136°}{2} = 0,375$$

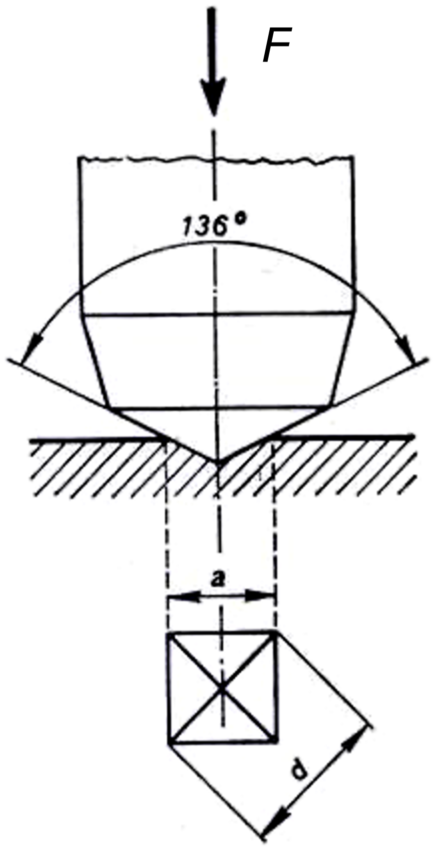
Admissible Test $$\frac{D_i}{D_b} \in [0,25\ ;\ 0,5]$$



Vickers Hardness Test

Widely used for testing metals and nonmetals of medium to high hardness.

A pyramid-shaped diamond indenter (136°) is pressed into specimen surface with a load (294 N = 30 kg) for 15 s.



$$HV = 0{,}102\frac{F}{S}$$

$$S = \frac{d^2}{2\sin\frac{136}{2}} = \frac{d^2}{1{,}854}$$

$$HV = 0{,}102 \cdot 1{,}854\frac{F}{d^2} = 0{,}1891\frac{F}{d^2}$$

HV = Vickers Hardness Number (VHN)
F = indentation load [N]
d = measured indentation diagonal [mm]

Rockwell Hardness Tests

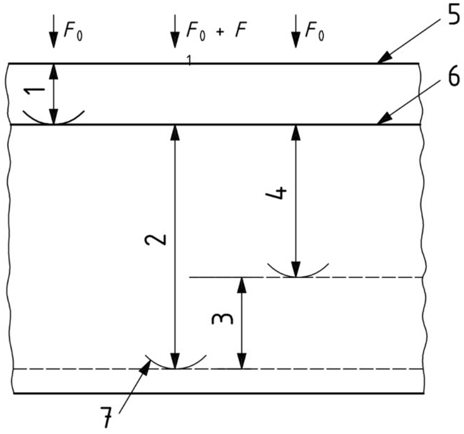
Forcing an indenter of specified size, shape and material into the surface of a test piece in two steps under specified conditions.

Measuring the permanent depth *h* of indentation under preliminary test force after removal of additional test force.

From the values of *h* and that of the two constants *N* and *S*, the Rockwell hardness (*HR#*) is calculated according to the formula:

$$HR\# = N - \frac{h}{S}$$

Rockwell Hardness Test



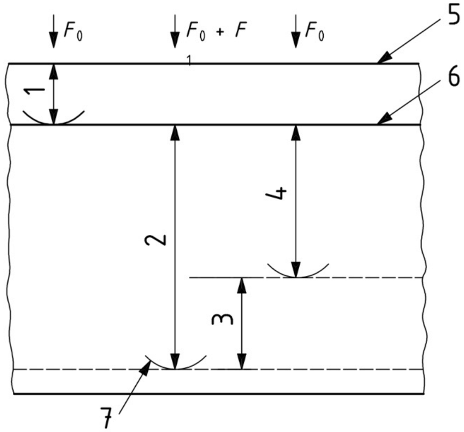
Key

1 Indentation depth by preliminary force F_0
2 Indentation depth by additional test force F_1
3 Elastic recovery just after removal of additional test force F_1
4 Permanent indentation depth h
5 Surface of specimen
6 Reference plane for measurement
7 Position of indenter

Symbol/ Abbreviated term	Designation	Unit
F_0	Preliminary test force	N
F_1	Additional test force	N
F	Total test force	N
S	Scale unit, specific to the scale	mm
N	Number, specific to the scale	
h	Permanent depth of indentation under preliminary test force after removal of additional test force (permanent indentation depth)	mm
HRA HRC HRD	Rockwell hardness $= 100 - \dfrac{h}{0{,}002}$	
HRB HRE HRF HRG HRH HRK	Rockwell hardness $= 130 - \dfrac{h}{0{,}002}$	
HRN HRT	Rockwell hardness $= 100 - \dfrac{h}{0{,}001}$	

Source: ISO 6508-1:2005

Rockwell Hardness Test



Key

1 Indentation depth by preliminary force F_0
2 Indentation depth by additional test force F_1
3 Elastic recovery just after removal of additional test force F_1
4 Permanent indentation depth h
5 Surface of specimen
6 Reference plane for measurement
7 Position of indenter

Rockwell hardness scale	Hardness symbol	Type of indenter	Preliminary test force F_0	Additional test force F_1	Total test force F	Field of application (Rockwell hardness test)
A[a]	HRA	Diamond cone	98,07 N	490,3 N	588,4 N	20 HRA to 88 HRA
B[b]	HRB	Ball 1,587 5 mm	98,07 N	882,6 N	980,7 N	20 HRB to 100 HRB
C[c]	HRC	Diamond cone	98,07 N	1,373 kN	1,471 kN	20 HRC to 70 HRC
D	HRD	Diamond cone	98,07 N	882,6 N	980,7 N	40 HRD to 77 HRD
E	HRE	Ball 3,175 mm	98,07 N	882,6 N	980,7 N	70 HRE to 100 HRE
F	HRF	Ball 1,587 5 mm	98,07 N	490,3 N	588,4 N	60 HRF to 100 HRF
G	HRG	Ball 1,587 5 mm	98,07 N	1,373 kN	1,471 kN	30 HRG to 94 HRG
H	HRH	Ball 3,175 mm	98,07 N	490,3 N	588,4 N	80 HRH to 100 HRH
K	HRK	Ball 3,175 mm	98,07 N	1,373 kN	1,471 kN	40 HRK to 100 HRK
15N	HR15N	Diamond cone	29,42 N	117,7 N	147,1 N	70 HR15N to 94 HR15N
30N	HR30N	Diamond cone	29,42 N	264,8 N	294,2 N	42 HR30N to 86 HR30N
45N	HR45N	Diamond cone	29,42 N	411,9 N	441,3 N	20 HR45N to 77 HR45N
15T	HR15T	Ball 1,587 5 mm	29,42 N	117,7 N	147,1 N	67 HR15T to 93 HR15T
30T	HR30T	Ball 1,587 5 mm	29,42 N	264,8 N	294,2 N	29 HR30T to 82 HR30T
45T	HR45T	Ball 1,587 5 mm	29,42 N	411,9 N	441,3 N	10 HR45T to 72 HR45T

[a] The field of application can be extended to 94 HRA for testing carbides.

[b] The field of application can be extended to 10 HRBW if specified in the product specification or by special agreement.

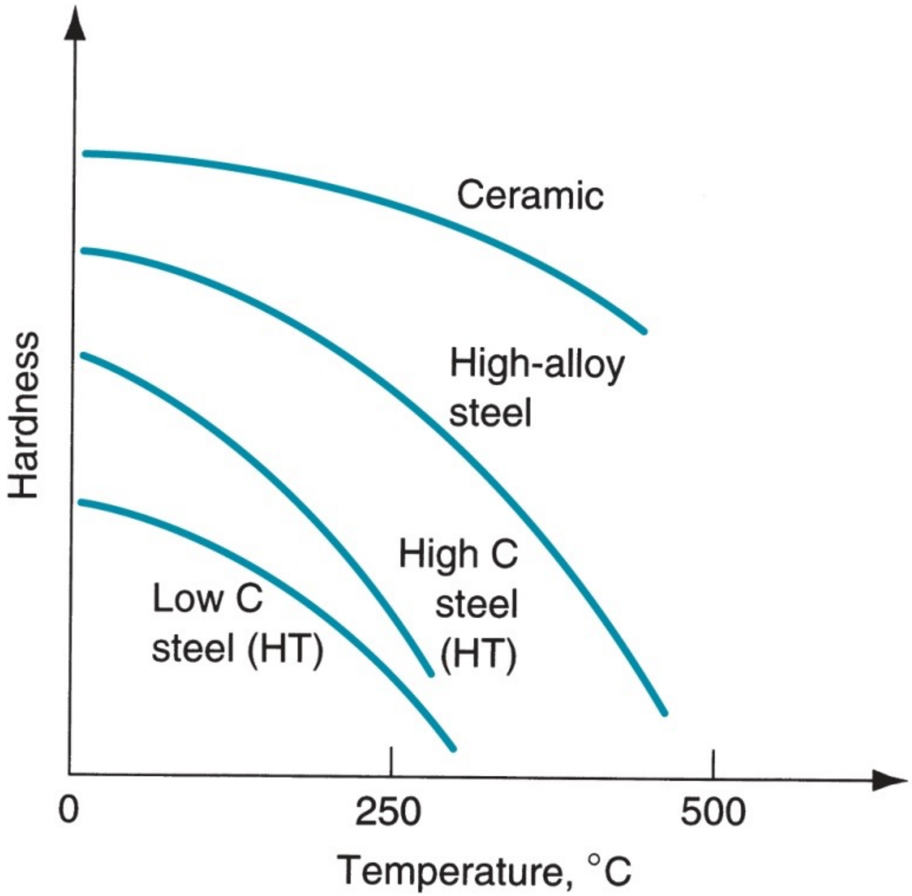
[c] The field of application can be extended to 10 HRC if the indenter possesses the appropriate dimensions.

NOTE Indenter balls with diameter 6,350 mm and 12,70 mm may also be used, if specified in the product specification or by special agreement.

Source: ISO 6508-1:2005

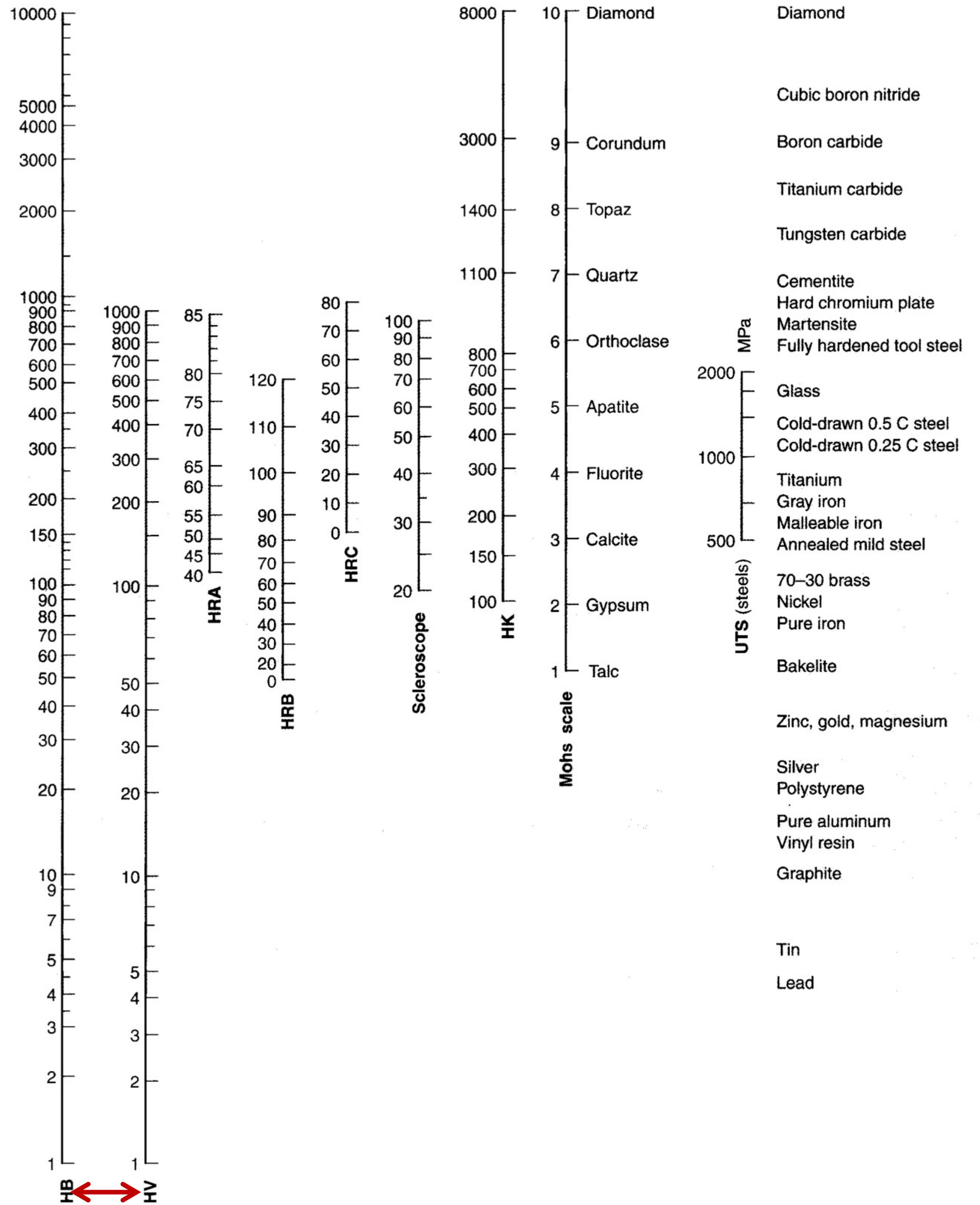
Hot Hardness

Ability of a material to retain hardness at elevated temperatures.



Typical hardness as a function of temperature for several materials.

HARDNESS SCALES



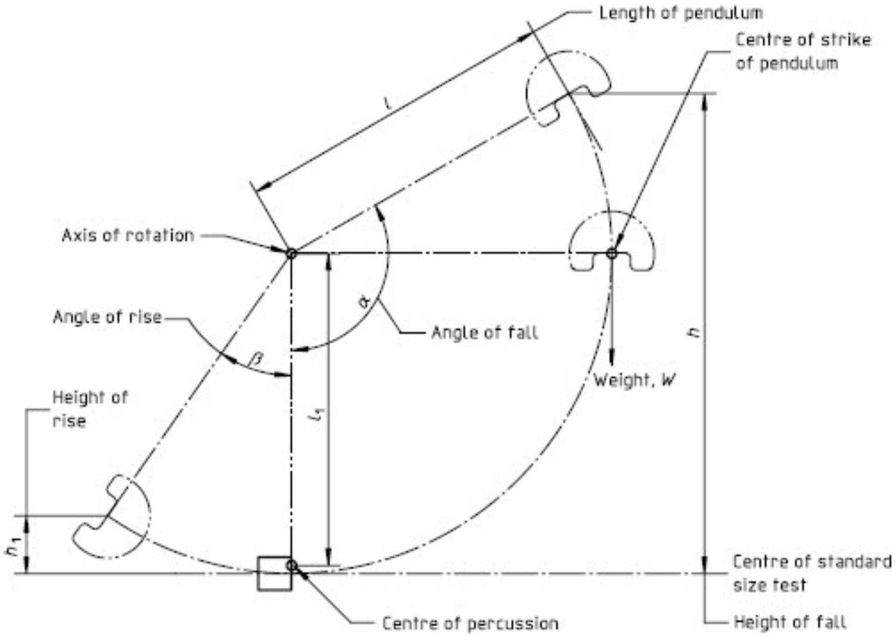
IMPACT TEST

Ability of a material to resist to impact.

A typical impact test consists of placing a notched specimen in an impact tester and breaking the specimen with a swinging pendulum.

From the amount of swing of the pendulum, the energy dissipated in breaking the specimen is obtained; this energy is the **impact toughness (K)** of the material.

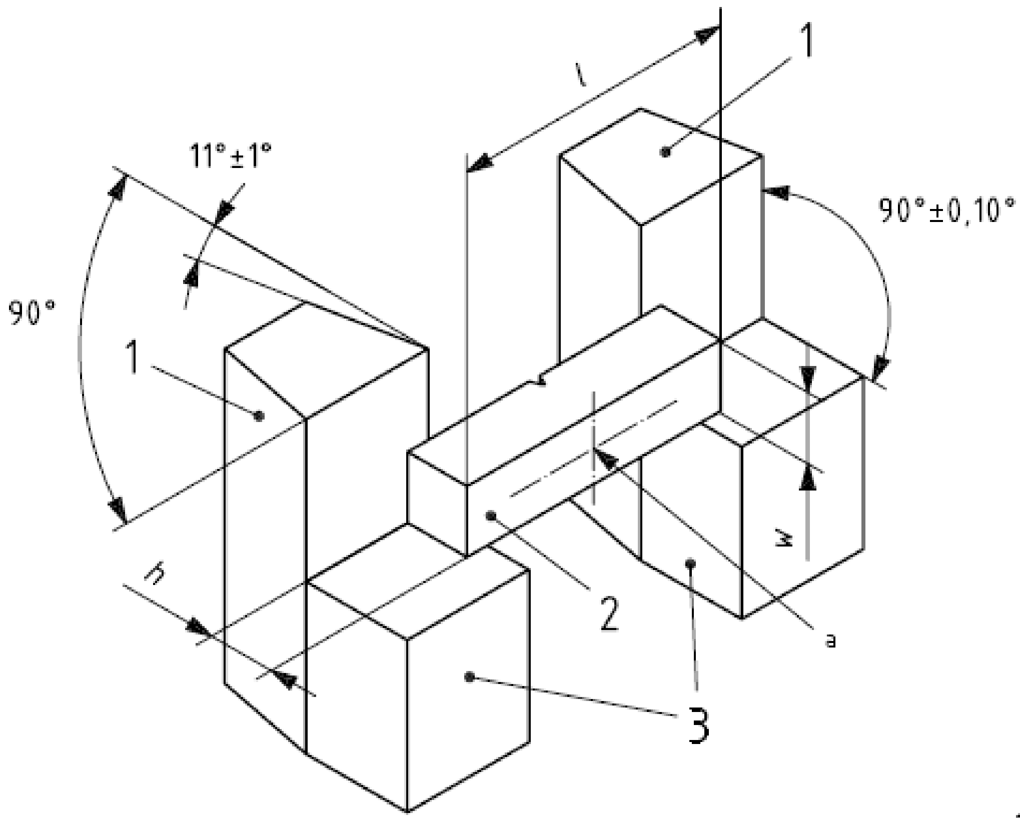
Charpy Test

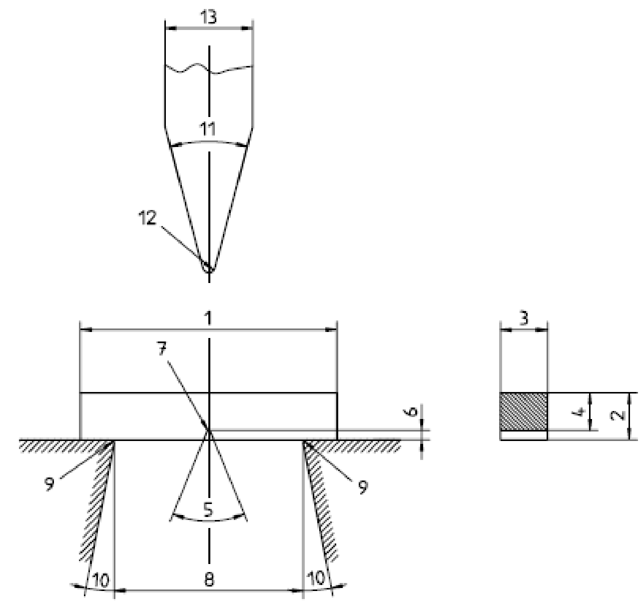


$$K = mg\,(h - h_1)\ \ [J]$$

$$v = \sqrt{2gl(1 - \cos\alpha)}$$

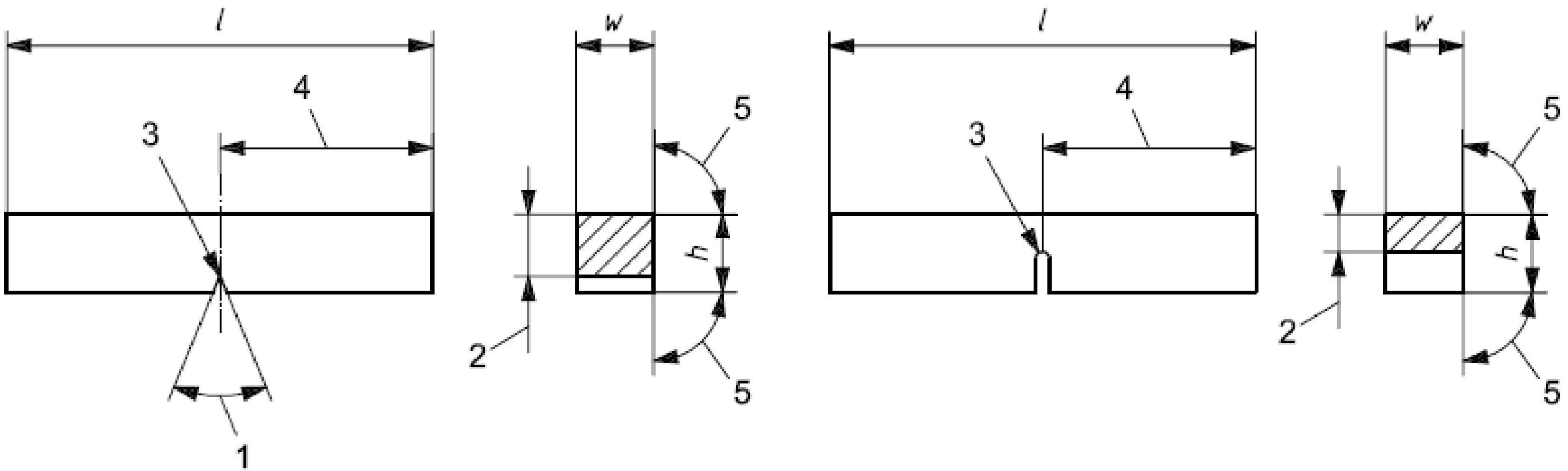
IMPACT TEST SETUP





1) 55 mm
2) 10 mm
3) 10 mm
8) 40 mm
11) 30°
12) R = 2 mm

IMPACT TEST SETUP



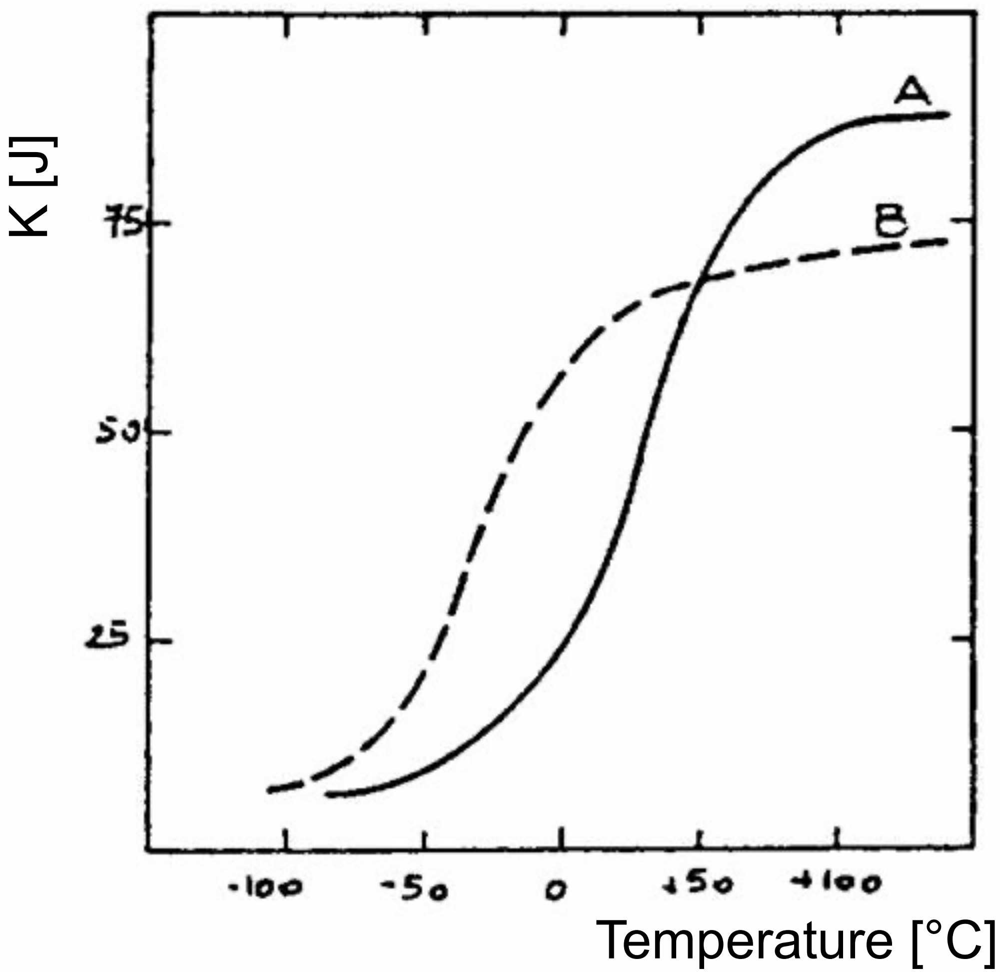
a) V-Notch Geometry

1) 45°
2) 10-1=9 mm
3) R = 0,25 mm

b) U-Notch Geometry

2) 10-5=5 mm
3) R = 1 mm

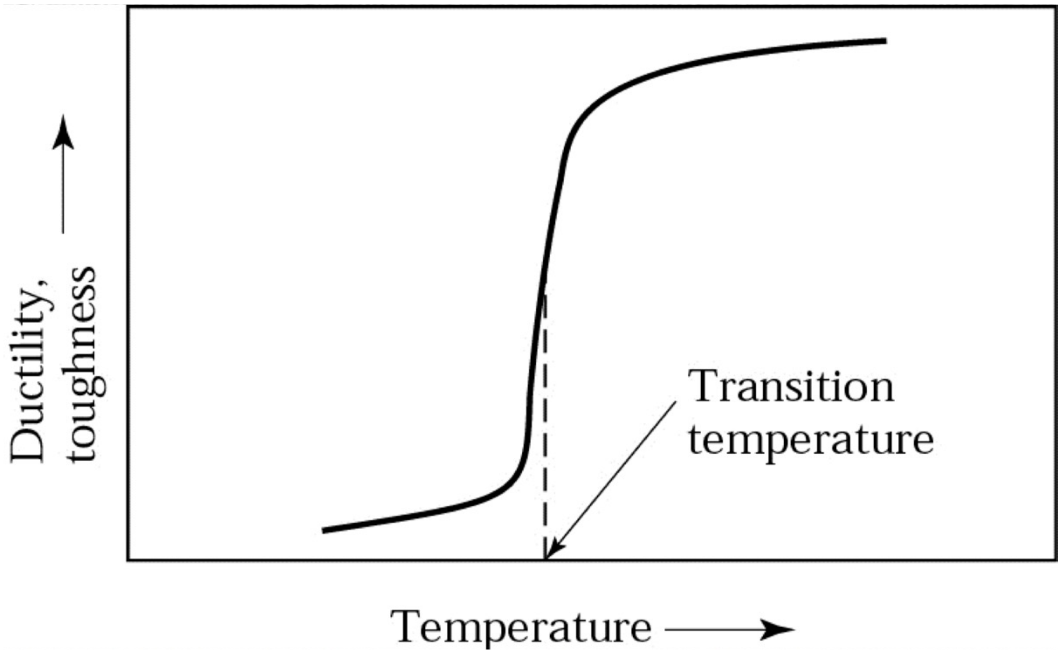
IMPACT TEST RESULT



Behaviour of two different steels.

Impact Test Result

Many metals undergo a sharp change in ductility and toughness across a narrow temperature range, called the **transition temperature**.

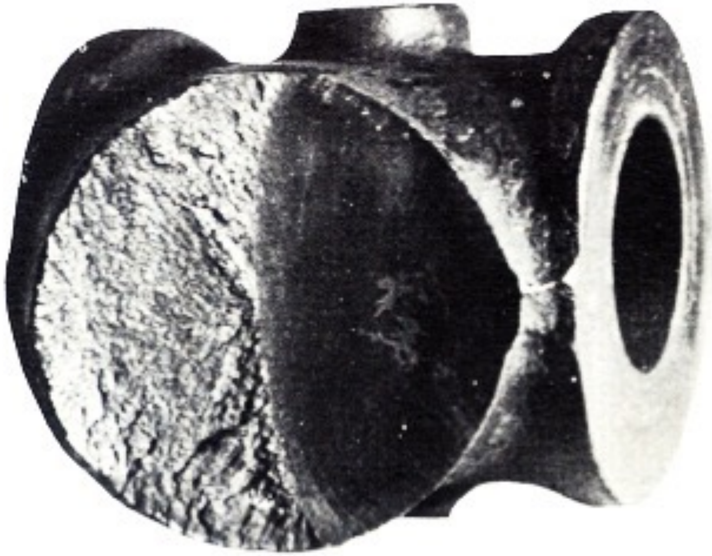


FATIGUE

Various components in a mechanical product are subject to rapidly fluctuating (cyclic or periodic) loads in addition to static loads.

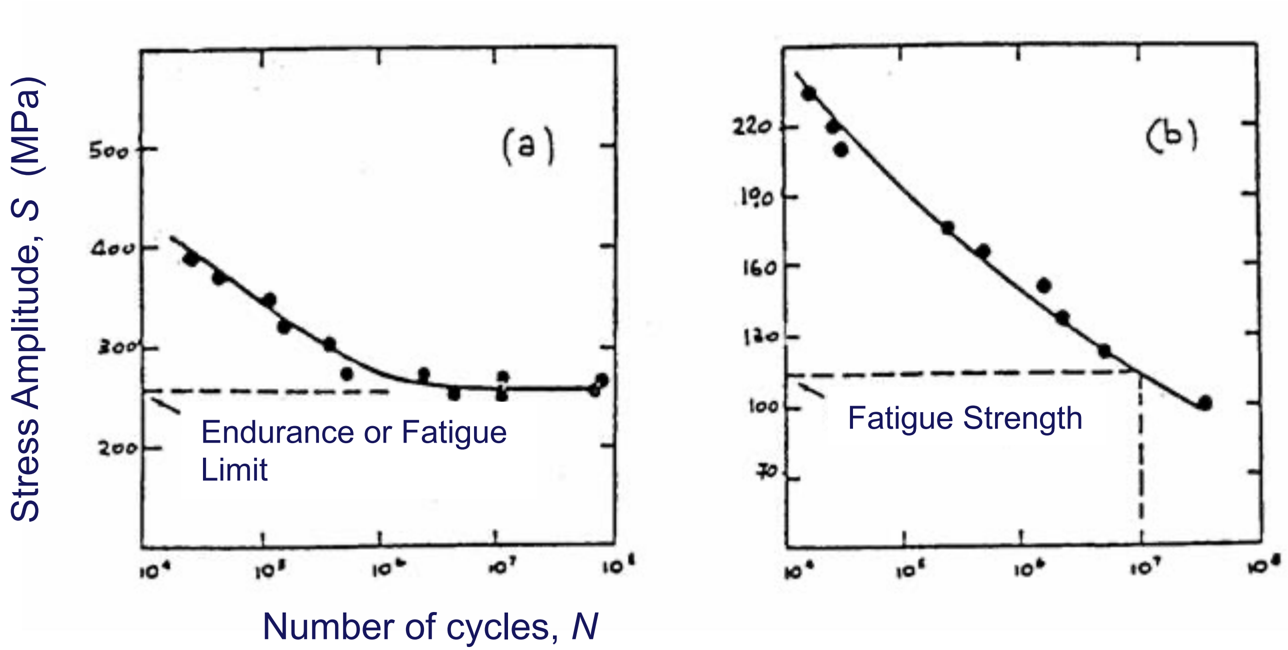
Under this conditions a component fails at a stress level below that at which failure would occur under static loading.

This failure is known as **fatigue failure**.



FATIGUE TEST

Fatigue test is carried out at various **stress amplitudes** (***S***); the **number of cycles** (***N***) it takes to cause total failure of the specimen is recorded.



SUMMARY

- The tensile test is the most commonly used test to determine mechanical properties of materials.
- From the tensile test it is possible to determine:
  - Modulus of elasticity (E)
  - Yield strength (Y)
  - Ultimate tensile strength (UTS)
  - Flow curve: strength coefficient (K) and strain hardening exponent (n)
  - Ductility and toughness.
- Compression, bending, shear tests give detailed mechanical properties, but these are related to tensile test results.
- Hardness, impact toughness and fatigue strength/limit are other very important mechanical properties of materials.